Pedro Henrique Ferreira da Silva

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SUMMARY

Master's student in Computer Science at the University of São Paulo (USP), holding a Bachelor's degree in Computer Science from the Federal University of Ceará (UFC). I have professional experience in software development and data science, having contributed to applications and intelligent solutions across the educational, legal, and industrial sectors. My current focus is on deep learning and artificial neural networks, particularly MLPs, CNNs, RNNs, GANs, and LLMs. My research interests include artificial intelligence, machine learning, computer vision, data science, combinatorial optimization, and software engineering.

Languages : Portuguese (Native), English (Advanced), French (Basic)
Tools : Git, GitHub, Docker, MySQL, PostgreSQL, SQLite, MongoDB, Linux
Programming Languages : Python, JavaScript, TypeScript, Java, PHP, C, C++
Libraries and Frameworks : PyTorch, TensorFlow, OpenCV, Scikit-Learn, Pandas, NumPy, React.js, Node.js
Soft Skills : Autonomy, Problem Solving, Teamwork, Adaptability

EXPERIENCE

LM Tech Sep. 2025 – Oct. 2025
Data Scientist (Freelance) Remote – Russas, CE, Brazil

- Developed a microservice for automated spreadsheet data correction.
- Applied data science techniques for data validation, standardization, and processing.
- Technologies: Python, Pandas, Flask

Cimento Apodi Oct. 2024 – Mar. 2025
Data Scientist (Intern) Russas, CE, Brazil

- Developed an intelligent solution for autonomous control of cement mills.
- Applied machine learning models to optimize industrial processes.
- Technologies: Pandas, NumPy, Scikit-Learn

Kolb Quintana Advogados Oct. 2022 – Dec. 2022
Software Developer (Freelance) Remote – São Paulo, SP, Brazil

- Developed a dashboard for automated document generation from spreadsheets.
- Implemented an automatic update mechanism with remote version checking.
- Technologies: TypeScript, React.js, Electron.js

Vivicell Oct. 2021 – Jan. 2022
Full Stack Developer (Intern) Morada Nova, CE, Brazil

- Developed a management system for mobile phone stores, including service order and warranty tracking.
- Implemented real-time communication between devices using Socket.IO.
- Technologies: React.js, Redux.js, Node.js, SQLite

DNA Evolutio Feb. 2021 – Aug. 2021
Full Stack Developer (Freelance) Morada Nova, CE, Brazil

- Developed a gamified educational platform for teaching cell biology.
- Implemented interactive quizzes, educational games, ranking systems, and multimedia content management.
- Technologies: React.js, Redux.js, Node.js, MySQL

EDUCATION

University of São Paulo (USP) 2026 – Present
M.Sc. in Computer Science São Paulo, SP, Brazil

Federal University of Ceará (UFC) 2022 – 2025
B.Sc. in Computer Science Russas, CE, Brazil

EEEP Osmira Eduardo de Castro 2019 – 2021
Technical Diploma in Information Technology Morada Nova, CE, Brazil

PUBLICATIONS

An ASP Approach for Academic Event Allocation: A Case Study at UFC 2026
XXIX Brazilian Symposium on Programming Languages (SBLP 2026) São Paulo, SP, Brazil

A Clustering-Based Max-SAT Hybrid Approach for the CVRP with Time Windows 2026
XI Theory of Computation Meeting (ETC 2026) Gramado, RS, Brazil

Post-improving the Capacitated Vehicle Routing Problem Using a Max-SAT Solver 2025
VI Brazilian Workshop on Logic (WBL 2025) Maceió, AL, Brazil

PROJECTS

JaguarLang – Language for Teaching and Research in Compilers 2025 – 2026
Development of a programming language focused on compiler education Russas, CE, Brazil

Fraud Detection in Financial Time Series 2025 – 2026
Application of LSTM networks to identify atypical patterns in financial data Russas, CE, Brazil

Machine Learning for Combinatorial Graph Problems 2023 – 2024
Application of supervised and unsupervised learning to NP-hard problems Russas, CE, Brazil

COURSES

Fundamentals of Deep Learning 2025
NVIDIA (8 hours) Russas, CE, Brazil

Machine Learning and Data Science with Python from A to Z 2024
Udemy (42 hours) Remote – São Paulo, SP, Brazil

Python 3 Course from Beginner to Advanced (with real projects) 2023
Udemy (124 hours) Remote – São Paulo, SP, Brazil

JavaScript and TypeScript Course from Beginner to Advanced 2022
Udemy (146 hours) Remote – São Paulo, SP, Brazil

Complete Web Development 20 courses + 20 projects 2020
Udemy (114 hours) Remote – São Paulo, SP, Brazil